

ENTRY 057b Reverse-Direction Confirmation (Sherbrooke → Kyiv) — August 25, 2026 — Status: COMPLETE

Entry 057b — Uncertainty Widening Holds in Both Cross-Chip Directions

Confirms the Asymmetry Seen Across Entries 051-056 is Reflected in Uncertainty, Not Just Error

Results: Both Directions

direction	warm MAE	warm std	cold MAE	cold std	widening ratio
Kyiv → Sherbrooke	0.90	0.57	4.63	1.57	2.77x
Sherbrooke → Kyiv	1.23	0.97	2.20	1.56	1.61x

A second independently-trained MC-Dropout model, trained on Sherbrooke only with Kyiv held out completely, reproduces the same qualitative pattern: uncertainty widens on the unseen chip (1.61x), correlating with a real error increase (MAE 1.23 to 2.20 pts). The widening is smaller than Kyiv-to-Sherbrooke's 2.77x -- consistent with every prior entry in this project finding Kyiv-to-Sherbrooke to be the harder transfer direction (Entries 047, 048, 051, 053-056). The uncertainty signal is not just detecting error, it is detecting the same directional asymmetry the point-prediction error metrics have shown all along, which is strong evidence the confidence estimate is tracking something real about the underlying transfer difficulty rather than being an artifact of one run.

Conclusion

Uncertainty quantification is validated in both directions and ready to expose in the live demo. The remaining work -- re-precomputing predictions with uncertainty for the full 16,002-pair lookup table used by the live demo -- is scoped separately given its size (comparable to Entry 046's original full-coverage precomputation).

Index Addendum (Entry 057b)

Scripts: entry057_mc_dropout.py (train_cc/eval_cc, direction argument added). Data: quantumbridge_data/entry057_mcdropout_cc_rev_results.json. Confirms 2.77x/1.61x asymmetric widening matches the point-prediction error asymmetry across the whole project.